# Donggyu **Joo**

Senior Machine Learning Scientist @ Samsung Research

jdg10126@gmail.com | donggyu-joo-a0a8ba22b | Donggyu Joo

*Potential O-1 Visa Candidate: 2,800+ Citations & 5+ Top-tier Publications*

## Summary

**Senior Machine Learning Scientist** with a **Ph.D. in AI** and 4+ years at **Samsung Research**, building on-device computer vision and vision-language AI for Samsung Galaxy devices. Shipped **8 on-device AI features** globally, including VLM-based semantic photo search for Galaxy Gallery with models under 100M parameters. Expertise spans large-scale multimodal data curation (50M+ image-text pairs), model training and evaluation, on-device optimization, and robustness improvement for real-world edge cases. **Team Lead** for 6 engineers, driving cross-functional execution across research, product, and engineering.

## Skills

| | |
|---|---|
| **Core AI & Research** | Computer Vision, Vision-Language Models, Image-Text Retrieval, Generative Modeling, Model Compression |
| **Production ML** | On-device AI, Model Optimization, Quantization, Distillation, TFLite Deployment, Edge-case Debugging |
| **Data & Evaluation** | Large-scale Data Curation, Model Training & Evaluation, Robustness Evaluation, Data Quality Control |
| **Languages & Tools** | Python, PyTorch, TensorFlow, TFLite, Git |

## Experience

**Samsung Research, Samsung Electronics** — *Seoul, S. Korea*

Senior Machine Learning Scientist — *Sep. 2022 - Present*

- **Gallery Photo Search & On-Device Visual Recognition:** Developed and deployed on-device multi-label image recognition model for Samsung Galaxy, enabling semantic photo search with models under 100M params. Expanded supported query coverage from 500 to 2,000+ categories, including documents, landmarks, and car brands while improving F1 score by 2%.
- **Large-scale Multimodal Data Curation:** Built GenAI-assisted labeling and filtering pipelines for 50M+ image-text pairs. Improved robustness by reducing noisy labels and increasing coverage of long-tail real-world image categories.
- **Research-to-Product:** Shipped 8 on-device AI models to production for Galaxy devices. Owned model optimization, TFLite conversion, on-device validation, latency tuning, and edge-case debugging based on user feedback.
- **Touch Sequence Classification:** Designed and optimized an on-device sequential classification model to detect unintended pocket touches. Achieved sensor-level reliability under challenging edge cases and deployed real-time CPU inference to Galaxy Z Flip6/7.
- **Technical Leadership:** Led 6 engineers across data curation, modeling, evaluation, and release. Collaborated with product and engineering stakeholders to turn ambiguous user-facing problems into deployable AI features.

## Education

**KOREA ADVANCED INSTITUTE OF SCIENCE AND TECHNOLOGY (KAIST)** — *Daejeon, S. Korea*

Ph.D. in Electrical Engineering — *Mar. 2018 - Aug. 2022*

- Thesis: Effective Deep Neural Network Compression Based on the Network Pruning
- Research Focus: Model Compression / Computer Vision

**KOREA ADVANCED INSTITUTE OF SCIENCE AND TECHNOLOGY (KAIST)** — *Daejeon, S. Korea*

M.S. in Electrical Engineering — *Sep. 2016 - Feb. 2018*

- Thesis: Unconstrained Control of Feature Map Size Using Non-integer Strided Sampling
- Research Focus: Generative Adversarial Networks (GANs) / Computer Vision

**KOREA ADVANCED INSTITUTE OF SCIENCE AND TECHNOLOGY (KAIST)** — *Daejeon, S. Korea*

B.S. in Electrical Engineering (Summa Cum Laude) — *Feb. 2013 - Aug. 2016*

# Publications [Citations: 2,800+ | h-index: 8]

## Selected Publications

- **Donggyu Joo**, Eojindl Yi, Sunghyun Baek, and Junmo Kim, "Linearly Replaceable Filters for Deep Network Channel Pruning", The Thirty-Fifth AAAI Conference on Artificial Intelligence (**AAAI**), 2021

- Janghyeon Lee, Hyeong Gwon Hong, **Donggyu Joo**, and Junmo Kim, "Continual Learning with Extended Kronecker-factored Approximate Curvature", IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2020

- Janghyeon Lee, **Donggyu Joo**, Hyeong Gwon Hong, and Junmo Kim, "Residual Continual Learning", Thirty-Fourth AAAI Conference on Artificial Intelligence (**AAAI**), Oral presentation, 2020

- **Donggyu Joo***, Doyeon Kim*, and Junmo Kim, "Generating a Fusion Image: One's Identity and Another's Shape", IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2018

- Junho Yim, **Donggyu Joo**, Jihoon Bae, and Junmo Kim, "A Gift From Knowledge Distillation: Fast Optimization, Network Minimization and Transfer Learning", IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2017

## Other Publications

- Doyeon Kim, Dongyeun Lee, **Donggyu Joo**, and Junmo Kim, "Training Cartoonization Network without Cartoon", 2023 IEEE International Conference on Image Processing (**ICIP**), 2023

- **Donggyu Joo**, Sunghyun Baek, and Junmo Kim, "Which Metrics for Network Pruning: Final Accuracy? Or Accuracy Drop?", The 29th IEEE International Conference on Image Processing (**ICIP**), 2022

- Doyeon Kim, Woonghyun Ka, Pyungwhan Ahn, **Donggyu Joo**, Sehwan Chun, and Junmo Kim, "Global-Local Path Networks for Monocular Depth Estimation with Vertical CutDepth", arXiv preprint arXiv:2201.07436, 2022

- **Donggyu Joo**, Doyeon Kim, Eojindl Yi, and Junmo Kim, "Linear Combination Approximation of Feature for Channel Pruning", Efficient Deep Learning for Computer Vision CVPR Workshop (**CVPRW**), 2022

- **Donggyu Joo**, Doyeon Kim, and Junmo Kim, "Slimming ResNet by Slimming Shortcut", 25th International Conference on Pattern Recognition (**ICPR**), 2020.

- Doyeon Kim, **Donggyu Joo**, and Junmo Kim, "Delivering Meaningful Representation for Monocular Depth Estimation", 25th International Conference on Pattern Recognition (**ICPR**), 2020.

- Doyeon Kim*, **Donggyu Joo***, and Junmo Kim, "TiVGAN: Text to Image to Video Generation with Step-by-Step Evolutionary Generator", **IEEE Access**, 2020.

- **Donggyu Joo**, Junho Yim, and Junmo Kim, "Unconstrained Control of Feature Map Size Using Non-integer Strided Sampling", 29th British Machine Vision Conference (**BMVC**), 2018

# Honors & Awards

2020 **Scholarship**, Young-Han Kim Global Leader Scholarship ($5,000)
2019 **Winner**, AI Reporter, AI WorldCup ($5,000)
2018 **Winner**, AI Reporter, AI WorldCup ($5,000)
2018 **Silver Prize**, 24th Samsung Humantech Paper Awards, Signal Processing ($7,000)
2017 **Silver Prize**, 23rd Samsung Humantech Paper Awards, Computer Science ($7,000)
2016 **Summa Cum Laude**, School of Electrical Engineering, KAIST
2015 **Dean's List**, School of Electrical Engineering, KAIST

# Patents

## United States

**Neural Network Learning Method and Device for Recognizing Class**

Junmo Kim, Byungju Kim, Juchang Kim, Yeakang Lee, Minsuk Park, Juseung Yun, Jaeyoung Lee, **Donggyu Joo**

U.S. Patent No. 11,449,733 B2 (Granted)

**Electronic Device for Processing Image and Operating Method Thereof**

**Donggyu Joo**, Dongchan Kim, Jinyoung Hwang

U.S. Patent Application Publication No. 2025/0292376 A1

## Republic of Korea

**Knowledge Transfer Method Using Deep Neural Network and Apparatus Therefor**

Junmo Kim, Junho Yim, **Donggyu Joo**

Republic of Korea Patent No. 10-2061935 (Granted)

**Knowledge Sharing Based Knowledge Transfer Method for Improving Quality of Knowledge and Apparatus Therefor**

Junmo Kim, Junho Yim, **Donggyu Joo**, Jihoon Bae, Naesoo Kim, Chulsik Pyo

Republic of Korea Patent No. 10-2063376 (Granted)

**Incremental Training Based Knowledge Transfer Method for Training Large Deep Neural Networks and Apparatus Therefor**

Junmo Kim, Junho Yim, **Donggyu Joo**, Jihoon Bae, Naesoo Kim, Chulsik Pyo

Republic of Korea Patent No. 10-2063377 (Granted)

**Electronic Device and Operating Method for the Same**

Sangwon Lee, Sahngmin Yoo, **Donggyu Joo**, Jiman Kim, Jinyoung Hwang

Republic of Korea Patent Application No. 10-2024-0114486

**Electronic Device Processing Image and Method of Operating the Same**

**Donggyu Joo**, Dongchan Kim, Jinyoung Hwang

Republic of Korea Patent Application No. 10-2024-0088511

**Electronic Apparatus for Identifying Abnormal Touch and Control Method Thereof**

**Donggyu Joo**, Jinyoung Hwang

Republic of Korea Patent Application No. 10-2024-0009651